Homework 5 - KNN

DATA 627 - Statistical Machine Learning, Fall 2025

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1 Predicting a Grade

Do by hand. A student wants to predict their final grade for the Statistical Machine Learning course using the KNN algorithm with $K = 3$. Seven friends who took the course last year had the following mid-term test scores and grades.

a) Estimate the prediction error rate of the algorithm, using the validation-set method, with Friends 2, 3, 4, and 6 as training and Friends 1, 5, 7 as testing data.

Friend	T **1**	TR **2**	TR **3**	TR **4**	T **5**	TR **6**	T **7**	$\hat{Y}$
Midterm	90	88	83	78	85	89	93	
Final Grade	**A**	A	B	B	**A**	B	**A**	B → INCORREC
Test 1: 90 Dist		88 2	83 7			89 1		
Test 1		A	B			B		
Test 5: 85 Dist		88 3	83 2			89 4		B → INCORRECT
Test 5								
Test 7: 93 Dist		88 5	83 10			89 4		B → INCORRECT
Test 7								

$$\text{The Erro Rate} = \frac{\#\text{ Incorrect prediction}}{\text{Total coses}} = \frac{3}{3} = 1.$$

Hence, the estimated error rate using validation method is 1 or 100%; All predictions were incorrect.

b) Estimate the prediction error rate of the algorithm using the leave-one-out cross-validation method.

LOOCV

Friend	1	2	3	4	5	6	7	
Midterm	90	88	83	78	85	89	93	
Final Grade	A	A	B	B	A	B	A	$\hat{Y}$
Dist - 1	X ↓	✓ 2	A2 vs 1B			✓ 1	✓ 3	A → SAME
Pred - 1								
Dist - 2	2	X ↓	5	10	3	1	5	
Pred - 2	A	A			A	B		A → SAME
Dist - 3	7	5	X ↓	5	2	6	10	
Pred - 3		A	B	B	A			A → INCORREC
Dist - 4	12	10	5	X ↓	7	11	15	
Pred - 4		A	B	B	A			A → INCORREC
Dist - 5	5	3	2	7	X ↓	4	8	
Pred - 5		A	B		A	B		B → INCORREC
Dist - 6	1	1	6	11	4	X ↓	4	
Pred - 6	A	A			A	B		A → INCORREC
Dist - 7	3	5	10	15	8	4	X ↓	
Pred - 7	A	A				B	A	A → SAME
Prediction	Error Rate $= \frac{4}{7} = 0.571$							

The estimated prediction error rate using LOOCV is 0.571 or 57.10%